

recall that a positive integer n and two integers a, b , $\gcd(a, n) = \gcd(b, n)$ if and only if a and b generate the same ideal in the ring $\mathbb{Z}/n$. This is also equivalent to the condition that $a \equiv ub \pmod{n}$ for some $u \in (\mathbb{Z}/n)^\times$. This property generalizes well for Artinian rings, and in particular, to finite rings. More precisely

Lemma 2.2. (See [5, Lemma 2.1]) *Let R be an Artinian ring. If $Ra = Rb$, then there exists $u \in R^\times$ such that $b = ua$.*

From this perspective, a crucial observation here is that a generating set S in a finite ring R such as $\mathbb{Z}/n$ or $\mathbb{F}_q[x]/f$ gives rise to a gcd-graph if and only if S is stable under the action of $R^\times$; that is, if $s \in S$, then $us \in S$ for all $u \in R^\times$. Motivated by this observation, we introduce the following definition.

Definition 2.3. We say that $\Gamma(R, S)$ is a gcd-graph if S is stable under the action of $R^\times$.

Remark 2.4. When $S = R^\times$, the graph $\Gamma(R, R^\times)$ is known as a unitary Cayley graph (see [1, 6]). This shows that the unitary Cayley graph over R is a special case of gcd-graphs. Other examples, as explained previously, include the gcd-graphs defined in [6] for the ring $\mathbb{Z}/n$ and the gcd-graphs defined in [9] for the ring $\mathbb{F}_q[x]/f$ where $\mathbb{F}_q$ is a finite field with q elements and f is a non-zero polynomial in $\mathbb{F}_q[x]$.

We provide below the necessary and sufficient conditions for a Cayley graph over R to be a gcd-graph. These conditions are somewhat more explicit than Definition 2.3. Furthermore, together with Corollary 2.7, this description will be important later on when we study the spectrum of these gcd-graphs.

Proposition 2.5. *Let R be a finite commutative ring and S a subset of R . Then the following are equivalent.*

- (1) $\Gamma(R, S)$ is a gcd-graph.
- (2) *There exist distinct nonzero principal ideals $\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_k$ such that for each $r \in R$, $r \in S$ if and only if there exists $1 \leq i \leq k$ such that $\mathcal{I}_i = Rr$.*

Proof. First, we claim that (2) $\implies$ (1). Clearly, S is symmetric and $0 \notin S$. By definition, we need to show that if $s \in S$ and $u \in R^\times$ then $us \in S$. Since $s \in S$, there exists an ideal $\mathcal{I}_i$ with $1 \leq i \leq k$ such that $Rs = \mathcal{I}_i$. Since $u \in R^\times$, $\mathcal{I}_i = Rs = Rus$. By (2), this shows that $us \in S$.

Let us show that (1) implies (2) as well. For each $s \in S$, Let $\mathcal{I}_s = Rs$ be the ideal generated by s . Let $\{\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_k\}$ be the set of all $\mathcal{I}_s$. We conclude that for each $s \in S$, $\mathcal{I}_s = \mathcal{I}_i$ for some $1 \leq i \leq k$. Conversely, if $r \in R$ such that $Rr \in \{\mathcal{I}_1, \mathcal{I}_2, \dots, \mathcal{I}_k\}$, then $Rr = Rs$ for some $s \in S$. By Lemma 2.2, we know that $r = us$ for some $u \in R^\times$. Since $\Gamma(R, S)$ is a gcd-graph, S is stable under the action of $R^\times$. This shows that $r \in S$ as well. $\square$